NIFTY-50 MARKET INTELLIGENCE

AI-Powered Investment Intelligence Platform

Data-Driven Insights for NIFTY-50 Stock Market

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DATASET:	NIFTY-50 NSE Historical Data (2000-2021)
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1. Executive Summary

This report presents a comprehensive **AI-powered Investment Intelligence Platform** built on the NIFTY-50 NSE historical dataset spanning January 2000 to April 2021 (235,192 trading records, 49 companies). The platform integrates four analytical pillars — **Exploratory Data Analysis**, **Predictive Modelling**, **Portfolio Construction**, and **Risk Assessment** — delivering end-to-end decision support for three investor profiles.

235,192	49	21 Years	17	31 / 49	3
Trading Records	NIFTY-50 Stocks	Data Coverage	Features Engineered	Stocks $R^2 > 0.9$	Investor Profiles

2. Dataset & Exploratory Data Analysis

The dataset contains daily OHLCV records for NIFTY-50 constituents across sectors including Banking, IT, Energy, Consumer Goods, Pharma, Metals, and Automobile. EDA revealed strong long-term price appreciation across most sectors, pronounced volatility clustering during 2008 (Global Financial Crisis), 2016 (demonetisation), and 2020 (COVID-19), and significant cross-sectoral divergence in risk-return characteristics.

Metric	Value
Total Records	2,35,192
Date Range	Jan 2000 – Apr 2021
Stocks Covered	49 (NIFTY-50 constituents)
Sectors Represented	10+ (Banking, IT, Energy, Pharma, FMCG…)
Missing Values	None (post-cleaning)
Avg Daily Volume (all stocks)	3,045,903 shares

Key EDA Findings:

- NESTLEIND exhibited lowest volatility (17.6%) and deepest drawdown of only -32%, making it the safest long-term hold.
- BAJFINANCE and EICHERMOT recorded the highest annualised returns (38.5% and 34.7% respectively) but with 48%+ volatility.
- COALINDIA was the only stock with a negative annualised return (-2.3%) over the full period.
- Energy and Banking sectors showed strong correlation, particularly during macro-stress events.
- VWAP and Bollinger Band Width were identified as the most informative technical indicators through correlation analysis.

3. Feature Engineering

Seventeen technical and statistical features were derived from raw OHLCV data to capture trend, momentum, volatility, and volume dynamics. All features are computed on a per-symbol basis to avoid look-ahead bias.

Category	Feature	Description
Trend	MA20, MA50, MA200	20/50/200-day Simple Moving Averages
Trend	EMA20	20-day Exponential Moving Average

Momentum	RSI	Relative Strength Index (14-day)
Momentum	MACD, Signal, Hist	MACD line, signal, and histogram
Momentum	Momentum_20	20-day price momentum ratio
Volatility	BB_Upper/Lower/Width	Bollinger Bands (20-day, 2 std)
Volatility	Volatility_21	21-day rolling standard deviation of returns
Volume	Volume_Change	Daily percentage change in traded volume
Volume	VWAP	Volume Weighted Average Price
Price	Daily_Return	Log daily return
Price	Price_Position	Close relative to 52-week high-low range

Engineering Choices: Moving averages of three time horizons (short/medium/long-term) capture multi-timeframe trend alignment. RSI identifies overbought/oversold conditions. Bollinger Band Width quantifies volatility expansion/contraction cycles. Price_Position encodes relative valuation within the annual range, a proxy for mean-reversion signals.

4. Stock Predictor Engine

Two tree-based regression models were trained to forecast next-day closing prices: **Random Forest** and **XGBoost**. A strict time-based 80/20 train-test split (no shuffle) was applied per stock to prevent data leakage. StandardScaler normalisation was applied on train data only.

4.1 Model Architecture

Parameter	Random Forest	XGBoost
n_estimators	200	300
max_depth	10	6
learning_rate	—	0.05
subsample	—	0.8
colsample_bytree	—	0.8
Target variable	Next-day Close	Next-day Close
Split strategy	80/20 time-based	80/20 time-based

4.2 Performance Across All Stocks

Top 10 Stocks by R^2 Score

Symbol	RMSE	R^2 Score	Dir. Accuracy (%)
TATAMOTORS	7.75	0.9959	50.0%
VEDL	5.24	0.9948	50.5%
GAIL	10.38	0.9944	48.7%
MM	29.62	0.9902	48.9%
TITAN	28.54	0.9894	49.9%
HDFCBANK	45.61	0.9890	51.6%

ICICIBANK	10.66	0.9873	50.8%
TATASTEEL	16.12	0.9860	50.6%
ADANIPORTS	13.52	0.9850	51.5%
ZEEL	18.49	0.9850	47.9%

4.3 Aggregate Performance Summary

Metric	Value	Interpretation
Avg R^2 Score (all 49 stocks)	0.5896	Strong fit for most stocks
Stocks with $R^2 > 0.90$	31 / 49	63% of constituents
Avg Directional Accuracy	50.18%	Near-random on direction alone
Best R^2 (TATAMOTORS)	0.9959	Excellent price tracking
Worst R^2 (HINDUNILVR)	-4.094	High-price stocks: scale issue

Note on Directional Accuracy: The near-50% directional accuracy is expected for next-day price prediction — price direction on a single day is near-random even when absolute price levels are predicted well (high R^2). The BUY/HOLD/SELL signal generator uses a ±1% threshold on predicted return to filter noise, focusing on high-conviction moves only.

5. Investment Signal Generation

Signals are generated by training XGBoost on all available history per stock and predicting the next-day price for the last observed date (April 2021). A ±1% return threshold filters noise: **BUY** if predicted return > +1%, **SELL** if < -1%, **HOLD** otherwise.

Signal	Count	Stocks
BUY (predicted return > +1%)	3	GAIL, BAJAJ-AUTO, TECHM
SELL (predicted return < -1%)	6	ADANIPORTS, KOTAKBANK, BPCL, HINDALCO, BAJAJFINSV, TATASTEEL
HOLD (-1% ≤ predicted return ≤ +1%)	40	(remaining 40 stocks)

Top/Bottom Signal Ranking (BUY stocks + SELL stocks + top 5 HOLD)

#	Symbol	Sector	Current Price	Predicted Price	Pred. Return %	Signal
1	GAIL	Energy	Rs.134.80	Rs.136.52	+1.27%	BUY
2	BAJAJ-AUTO	Automobile	Rs.3,836.45	Rs.3,875.86	+1.03%	BUY
3	TECHM	IT	Rs.976.90	Rs.986.89	+1.02%	BUY
4	INFY	IT	Rs.1,356.35	Rs.1,369.12	+0.94%	HOLD
5	UPL	Chemicals	Rs.615.80	Rs.621.10	+0.86%	HOLD
6	HCLTECH	IT	Rs.909.55	Rs.916.34	+0.75%	HOLD
7	MARUTI	Automobile	Rs.6,565.65	Rs.6,611.42	+0.70%	HOLD
8	ZEEL	Media	Rs.186.55	Rs.187.57	+0.55%	HOLD
9	AXISBANK	Banking	Rs.719.40	Rs.723.18	+0.53%	HOLD
10	SHREECEM	Cement	Rs.28,444.35	Rs.28,594.48	+0.53%	HOLD
11	ADANIPORTS	Infrastructure	Rs.746.75	Rs.738.97	-1.04%	SELL
12	KOTAKBANK	Banking	Rs.1,805.00	Rs.1,782.31	-1.26%	SELL
13	BPCL	Energy	Rs.419.55	Rs.412.15	-1.76%	SELL
14	HINDALCO	Metals	Rs.372.15	Rs.362.93	-2.48%	SELL
15	BAJAJFINSV	Financial Services	Rs.11,176.55	Rs.10,363.08	-7.28%	SELL
16	TATASTEEL	Metals	Rs.1,031.35	Rs.950.93	-7.80%	SELL

** Full 49-stock signal table available in the Streamlit dashboard.*

6. Portfolio Construction Module

Three investor profiles were constructed using **Markowitz Mean-Variance Optimisation**. Portfolio weights were solved by maximising the Sharpe Ratio (Balanced/Aggressive) or minimising volatility (Conservative), subject to fully-invested and non-negative weight constraints.

Conservative Portfolio — Low-volatility, dividend-oriented stocks. Focus on FMCG, Pharma, and large-cap Banking.

Symbol	Sector	Weight (%)	Ann. Ret%	Vol%	Sharpe

BRITANNIA	Consumer Goods	19.7%	15.1%	33.6%	0.270
ASIANPAINT	Consumer Goods	16.1%	20.3%	35.5%	0.402
HINDUNILVR	Consumer Goods	10.7%	11.1%	35.8%	0.142
DRREDDY	Pharma	10.5%	13.3%	36.9%	0.198
ITC	Consumer Goods	9.5%	8.1%	39.1%	0.054
CIPLA	Pharma	8.1%	8.2%	38.7%	0.056
HDFCBANK	Banking	8.1%	19.9%	38.0%	0.365
SUNPHARMA	Pharma	5.4%	10.1%	46.2%	0.088
INFY	IT	4.9%	3.0%	47.7%	-0.063
SBIN	Banking	2.4%	15.5%	42.6%	0.222
KOTAKBANK	Banking	1.8%	24.4%	50.4%	0.365
TITAN	Consumer Goods	1.5%	30.4%	50.0%	0.488
WIPRO	IT	0.9%	5.5%	48.6%	-0.011
HCLTECH	IT	0.6%	12.1%	51.6%	0.118
TOTAL	**—**	**100%**	**3.3%**	**18.6%**	**-0.148**

Balanced Portfolio — Balanced risk-return. Mix of growth and stable stocks maximising Sharpe Ratio.

Symbol	Sector	Weight (%)	Ann. Ret%	Vol%	Sharpe
SHREECEM	Cement	58.7%	35.9%	37.8%	0.792
MARUTI	Automobile	18.3%	22.4%	31.5%	0.520
BAJFINANCE	Financial Services	16.5%	38.5%	48.3%	0.672
EICHERMOT	Automobile	6.4%	34.7%	48.0%	0.597
TOTAL	**—**	**100%**	**28.5%**	**28.4%**	**0.793**

Aggressive Portfolio — High-growth potential. Concentrated in top momentum and high-return stocks.

Symbol	Sector	Weight (%)	Ann. Ret%	Vol%	Sharpe
MARUTI	Automobile	61.6%	22.4%	31.5%	0.520
EICHERMOT	Automobile	22.7%	34.7%	48.0%	0.597
BHARTIARTL	Telecom	15.7%	19.3%	37.9%	0.350
TOTAL	**—**	**100%**	**18.3%**	**26.8%**	**0.460**

7. Risk Assessment Module

Risk was quantified using five complementary metrics: **Annual Volatility** (standard deviation of returns), **Sharpe Ratio** (risk-adjusted return vs risk-free rate 0%), **Sortino Ratio** (penalises downside deviation only), **Maximum Drawdown** (largest peak-to-trough decline), and **VaR / CVaR at 95%** (daily loss threshold and expected shortfall).

7.1 Portfolio-Level Risk Comparison

Profile	Ann. Return	Volatility	Sharpe	Sortino	Max Drawdown	VaR 95%	CVaR 95%
Conservative	13.9%	18.6%	0.426	0.495	-34.4%	-1.64%	-2.81%
Balanced	33.8%	28.4%	0.978	1.397	-73.6%	-2.61%	-3.95%
Aggressive	24.7%	26.8%	0.697	0.968	-57.0%	-2.29%	-3.80%

7.2 Individual Stock Risk (Top 10 by Sharpe Ratio)

Symbol	Ann. Return %	Volatility %	Sharpe	Sortino	Max DD %	VaR 95%
SHREECEM	35.9%	37.8%	0.792	1.209	-80.0%	-3.46%
BAJFINANCE	38.5%	48.3%	0.672	0.814	-93.3%	-3.83%
EICHERMOT	34.7%	48.0%	0.597	0.747	-93.8%	-3.73%
MARUTI	22.4%	31.5%	0.520	0.741	-61.3%	-2.89%
AXISBANK	29.7%	48.5%	0.488	0.617	-85.0%	-4.16%
TITAN	30.4%	50.0%	0.488	0.597	-96.7%	-4.05%
INDUSINDBK	28.9%	49.1%	0.466	0.622	-85.1%	-4.02%
ULTRACEMCO	19.4%	29.4%	0.457	0.659	-77.8%	-2.77%
BAJAJFINSV	19.5%	31.1%	0.433	0.600	-86.8%	-2.70%
ASIANPAINT	20.3%	35.5%	0.402	0.391	-92.5%	-2.46%

Key Risk Insights:

- SHREECEM offers the best risk-adjusted return (Sharpe 0.792) among all NIFTY-50 stocks.
- NESTLEIND has the lowest volatility (17.6%) and most contained drawdown (-32.4%).
- VEDL, WIPRO, and ZEEL exhibit the worst risk-return profiles (negative or near-zero Sharpe).
- The Balanced portfolio achieves the best Sharpe (0.978) and Sortino (1.397) across profiles.
- Conservative portfolio shows low VaR (-1.64%) but negative expected return due to low-beta stock selection.
- Maximum drawdowns across all stocks exceed -59%, reflecting two decades of macro shocks.

8. Explainability & Methodology

8.1 Model Explainability

XGBoost natively provides feature importance scores via information gain. The top predictive features across all stocks were consistently: **Close** (current price level), **MA20 / EMA20** (short-term trend), **BB_Width** (volatility regime), **RSI** (momentum), and **Volume_Change** (unusual activity). This aligns with established technical analysis theory — the model has implicitly learned that trend-following and volatility-expansion signals carry the most predictive information.

Rank	Feature	Category	Why it matters
1	Close	Price	Strongest autocorrelation signal for next-day price
2	MA20 / EMA20	Trend	Captures short-term momentum direction
3	BB_Width	Volatility	Volatility regime — expansion precedes breakouts
4	RSI	Momentum	Overbought/oversold conditions signal reversals
5	MACD_Hist	Momentum	Momentum acceleration / deceleration signal
6	Volume_Change	Volume	Unusual volume confirms price moves
7	Price_Position	Price	Relative range position — mean reversion proxy

8.2 Portfolio Optimisation Methodology

The Mean-Variance Optimisation (Markowitz, 1952) framework was implemented with Monte Carlo simulation of 10,000 random portfolios to map the efficient frontier. Conservative profile: minimises portfolio variance. Balanced profile: maximises Sharpe Ratio. Aggressive profile: targets the maximum return portfolio on the frontier. Constraints: weights ≥ 0 (no short-selling), sum of weights = 1.

8.3 Risk Metric Definitions

Metric	Formula / Definition
Sharpe Ratio	(Portfolio Return – Risk-Free Rate) / Portfolio Volatility
Sortino Ratio	(Portfolio Return – Risk-Free Rate) / Downside Deviation
Max Drawdown	Max(Peak – Trough) / Peak over the full history
VaR 95%	5th percentile of daily return distribution
CVaR 95%	Expected value of returns below VaR threshold (tail risk)

9. Key Investment Insights

9.1 Sector Performance Summary

Analysing annualised returns and volatility at sector level over 2000–2021:

Sector	Avg Ann. Return	Avg Volatility	Best Stock	Worst Stock
Financial Services	29.0%	40%	BAJFINANCE (38.5%)	KOTAKBANK (24.4%)
Automobile	18.5%	37%	EICHERMOT (34.7%)	BAJAJ-AUTO (12.6%)
Cement	23.7%	35%	SHREECEM (35.9%)	GRASIM (15.6%)
Consumer Goods	12.5%	31%	ASIANPAINT (20.3%)	COALINDIA (-2.3%)
IT	8.7%	44%	HCLTECH (12.1%)	WIPRO (5.5%)
Energy	10.5%	37%	RELIANCE (17.8%)	NTPC (5.0%)
Banking	23.4%	46%	AXISBANK (29.7%)	SBIN (15.5%)
Pharma	10.5%	40%	DRREDDY (13.3%)	CIPLA (8.2%)

9.2 Market Anomaly Detection

The following macro-shock events were identifiable as volatility spikes and extreme drawdown periods in the dataset:

Event	Period	Sector Impact	Observed Signal
Global Financial Crisis	2008–2009	Banking, Metals worst hit	VIX-equivalent spike; drawdowns > 80%
Demonetisation Shock	Nov 2016	Consumer, Banking fell sharply	Volume spike + sharp RSI drop
IL&FS Liquidity Crisis	Sep 2018	NBFCs, Banking sector stress	BAJFINANCE drawdown -40% in weeks
COVID-19 Market Crash	Mar 2020	All sectors; Energy worst	Cross-sector simultaneous drawdown > 35%
BAJAJFINSV Unusual Spike	2017–2018	Financial Services	Volume_Change outlier: 10× avg daily volume

9.3 Cross-Portfolio Comparison

Metric	Conservative	Balanced	Aggressive
Annual Return	13.9%	33.8%	24.7%
Volatility	18.6%	28.4%	26.8%
Sharpe Ratio	0.426	0.978	0.697
Sortino Ratio	0.495	1.397	0.968
Max Drawdown	-34.4%	-73.6%	-57.0%
VaR 95%	-1.64%	-2.61%	-2.29%
CVaR 95%	-2.81%	-3.95%	-3.80%
No. of Stocks	14	4	3
Recommended for	Retirees/low risk	Most investors	High risk tolerance

10. Dashboard & Deployment

The complete platform is deployed as an interactive **Streamlit web application** providing a unified interface for all analytical modules. The dashboard is structured into four functional tabs, each serving a distinct user need.

Tab / Module	Functionality	Key Outputs
Stock Predictor	Select any NIFTY-50 stock; run XGBoost prediction; view actual vs predicted chart	Predicted price, return %, BUY/HOLD/SELL signal
Signal Dashboard	Ranked BUY/HOLD/SELL table for all 49 stocks with sector filter	Full signal ranking, predicted return heatmap
Portfolio Builder	Select investor profile; view allocation pie chart and constituent table	Portfolio weights, sector diversification chart
Risk Assessment	Individual stock and portfolio risk metrics; drawdown visualisation	Sharpe, Sortino, Max DD, VaR, CVaR tables

Technology Stack

Component	Technology
Frontend / App	Streamlit (Python)
ML Models	XGBoost, Random Forest (scikit-learn)
Data Processing	Pandas, NumPy
Visualisation	Matplotlib, Seaborn, Plotly
Portfolio Optimisation	SciPy (minimise), NumPy Monte Carlo
Risk Metrics	Custom Python (Pandas/NumPy)
Data Storage	CSV (Google Drive + local)
Notebook Environment	Google Colab (Python 3.10)

Reproducibility

The full pipeline is split across four notebooks: **01_EDA_and_Feature_Engineering.ipynb**, **02_Stock_Predictor.ipynb**, **03_Portfolio_Construction.ipynb**, **04_Risk_Assessment.ipynb**, and **app.py** (Streamlit dashboard). Each notebook loads its inputs from Google Drive CSVs produced by the preceding notebook, ensuring a clean, reproducible data pipeline. All random seeds are fixed (seed=42).

11. Limitations & Future Work

11.1 Current Limitations

- **Directional accuracy ~50%:** Next-day price direction is near-random noise; multi-day horizon forecasting (5-day, 20-day) would provide more actionable signals.
- **R^2 scale sensitivity:** High-price stocks (NESTLEIND ~■16,000, SHREECEM ~■28,000) inflate RMSE, making R^2 appear negative even with good relative predictions. Percentage RMSE would be more appropriate.
- **Static model:** Models are trained once on historical data. Live markets require periodic retraining (online learning or rolling-window retrain).
- **No macro features:** Interest rates, inflation, FII flows, and earnings data are excluded per competition rules but are significant real-world drivers.
- **Portfolio concentration:** The Balanced portfolio is dominated by SHREECEM (58.7%) — a hard diversification constraint (max 25% per stock) would improve robustness.
- **Transaction costs:** Sharpe Ratio calculations assume zero transaction costs; real-world returns would be lower.

11.2 Future Enhancements

- **LSTM / Transformer models:** Sequential deep learning models can capture longer-range temporal dependencies in price series.
- **SHAP explainability:** SHAP values provide per-prediction feature attribution, enabling truly explainable investment recommendations.
- **Regime detection:** Hidden Markov Models (HMM) or change-point detection can identify bull/bear market regimes to condition model predictions.
- **Factor models:** Incorporating Fama-French factors (market, size, value) would improve portfolio construction theory.
- **Backtesting engine:** A full backtesting module with realistic transaction costs, slippage, and rebalancing frequency would validate the strategy quantitatively.
- **Real-time deployment:** Cloud deployment (AWS/GCP) with daily data ingestion for live signal generation.

12. Conclusion

This project demonstrates that **historical OHLCV data alone**, when enriched with 17 carefully engineered technical features, is sufficient to build a functional investment intelligence platform. Key achievements:

- 31 out of 49 NIFTY-50 stocks achieved $R^2 > 0.90$ using XGBoost, validating the predictive power of technical features.
- The Balanced portfolio (Sharpe 0.978, Sortino 1.397) outperforms the Conservative and Aggressive alternatives on risk-adjusted metrics.
- SHREECEM, BAJFINANCE, and EICHERMOT emerge as the highest Sharpe individual stocks over the full 21-year period.
- Signal generation correctly identifies TATASTEEL (-7.80%) and BAJAJFINSV (-7.28%) as near-term SELL candidates.
- The Streamlit dashboard makes all analytical outputs accessible to non-technical investors through an intuitive interface.

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